**Shreve: Stochastic Calculus for Finance II**
**Key Results, Theorems, Definitions, etc.**

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**Abstract**

This document contains a summary of key results, theorems, and definitions that I deem important or interesting. This is not a comprehensive summary of the textbook, but a taste...

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# 1. General Probability Theory

## 1.1. Infinite Probability Spaces

Used to model a situation in which a random experiment with infinitely many possible outcomes is conducted. The problem with uncountably infinite sample spaces, like uniform sampling from the interval $[0, 1]$, is that for most experiments the probability of any particular outcome is 0.

**Definition 1** ($\sigma$-algebra)**.** Let $\Omega$ be a nonempty set, and $\mathcal{F}$ a collection of subsets of $\Omega$. We say that $\mathcal{F}$ is a **$\sigma$-algebra** if the following holds:

(1) $\emptyset \in \mathcal{F}$

(2) $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$

(3) $A_1, A_2, \dots \in \mathcal{F} \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$

Note that this also implies closed under finite unions and closed under countable intersections (since intersection is complement of union of complements). Additionally the whole space $\Omega = \emptyset^c \in \mathcal{F}$.

**Definition 2** (probability measure)**.** Let $\Omega$ nonempty set, $\mathcal{F}$ a $\sigma$-algebra of subsets of $\Omega$. A **probability measure** $\mathbb{P}$ is a function that assigns a number in $[0, 1]$ to every set $A \in \mathcal{F}$ called the probability of $A$ and written $\mathbb{P}(A)$. We require that

(1) $\mathbb{P}(\Omega) = 1$

(2) (countable additivity) If $A_1, A_2, \ldots$ is a sequence of disjoint sets in $\mathcal{F}$, then

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n)$$

The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is called a **probability space**.

**Example 1.** *Uniform Lebesque Measure on* $[0,1]$

Define the probability of closed intervals $[a,b]$ to be $\mathbb{P}[a,b] = b-a$, $0 \le a \le b \le 1$. Note if $a = b$ then $[a,b] = \{a\}$ and has a probability of 0. Note this also implies $\mathbb{P}(a,b) = \mathbb{P}[a,b] = b-a$.

All sets whose probabilities are determined by the defined formula and the properties of probability measures are exactly those belonging to the **Borel $\sigma$-algebra** $\mathcal{B}([0,1])$, which is the $\sigma$-algebra generated by the set of open or closed sub-intervals. The sets in this $\sigma$-algebra are called **Borel sets**.

**Example 2.** *Infinite, Independent Coin Toss Space*

A coin is tossed infinitely many times and we let $\Omega_\infty$ be the set of possible outcomes. The probability of a head is $p > 0$ and of tails $q = 1-p > 0$. We can define a probability measure that gives a probability to every set that can be described in terms of finitely many tosses, which in turn determines the probability of sets that are not describable with finitely many coin tosses. One can then show that every sequence in $\Omega_\infty$ has probability zero. This brings about the notion of *almost sure.*

**Definition 3** (almost sure)**.** in the probability space $(\Omega, \mathcal{F}, \mathbb{P})$, if a set $A \in \mathcal{F}$ satisfies $\mathbb{P}(A) = 1$ then it occurs **almost surely**.

### 1.2. Random Variables and Distributions

**Definition 4.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A **random variable** is a real valued function $X$ defined on $\Omega$ with the property that for every Borel set $B \in \mathcal{B}(\mathbb{R})$, the subset of $\Omega$ given by

$$\{X \in B\} = \{\omega \in \Omega : X(\omega) \in B\}$$

is in the $\sigma$-algebra $\mathcal{F}$.

**Example 3.** *Stock Prices*

Consider the space of infinite, independent coin tosses from before $(\Omega_\infty, \mathcal{F}_\infty, \mathbb{P})$. Define stock prices using the formulas

$$S_0(\omega) = 4,$$

$$S_1(\omega) = \begin{cases} 8, & \omega_1 = H, \\ 2, & \omega_1 = T, \end{cases}$$

$$S_2(\omega) = \begin{cases} 16, & \omega_1 = \omega_2 = H, \\ 4, & \omega_1 \neq \omega_2, \\ 1, & \omega_1 = \omega_2 = T, \end{cases}$$

$$S_{n+1}(\omega) = \begin{cases} 2S_n(\omega), & \omega_{n+1} = H, \\ \frac{1}{2}S_n(\omega), & \omega_{n+1} = T. \end{cases}$$

These are all random variables and thus we can compute the probability they take certain values, like $\mathbb{P}\{S_2 = 4\} = \mathbb{P}(A_{HT} \cup A_{TH}) = 2pq$.

**Definition 5** (distribution measure)**.** Let $X$ be a random variable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The **distribution measure** of $X$ is the probability measure $\mu_X$ that assigns to each Borel subset $B \subset \mathbb{R}$ the mass $\mu_X(B) = \mathbb{P}\{X \in B\}$.

It's important to note that random variables and distributions are two different concepts. Two different random variables can have the same distribution and a single random variable can have two different distributions.

**Example 4.** *Distributions*

Consider the uniform measure on $[0, 1]$ defined before, and call it $\mathbb{P}$. Now consider $X(\omega) = \omega$, and $Y(\omega) = 1-\omega$. These are two distinct RVs, both defined on $\Omega = [0, 1]$, but they share the same distribution:

$$\mu_Y[a, b] = \mathbb{P}\{Y \in [a, b]\} = \mathbb{P}\{\omega : a \leq 1 - \omega \leq b\} = \mathbb{P}[1 - b, 1 - a] = (1 - a) - (1 - b) = b - a = \mu_X[a, b]$$

Note we can also define a probability measure on $[0, 1]$ such that the distribution of $X$ and $Y$ are different. One such is example is the following:

$$\tilde{\mathbb{P}}[a, b] = \int_a^b 2\omega \, d\omega = b^2 - a^2, \quad 0 \leq a \leq b \leq 1$$

Once the distribution measure is defined for all closed intervals $[a, b] \subset \mathbb{R}$, it is determined for every Borel subset of $\mathbb{R}$.

We can also record the distribution of an RV in terms of its **cumulative distribution function** (cdf), which is defined as $F(x) = \mathbb{P}\{X \leq x\} = \mu_X(-\infty, x]$, $x \in \mathbb{R}$. The cdf and the distribution measure essentially give the same amount of information, but we can have more detail in two special cases. Firstly if there's a nonnegative **density function** $f(x)$ such that

$$\mu_X[a, b] = \mathbb{P}\{a \leq X \leq b\} = \int_a^b f(x) \, dx \quad -\infty < a \leq b < \infty$$

Second if there's a **probability mass function**. In this case there exists a possibly infinite sequence $x_1, x_2, \ldots$ such that the rv X takes a value in the sequence with probability one. In this case $p_i = \mathbb{P}\{X = x_i\}$ The mass assigned to a borel set $B \subset \mathbb{R}$ is

$$\mu_X(B) = \sum_{i : x_i \in B} p_i \quad B \in \mathcal{B}(\mathbb{R})$$

**Example 5.** *Standard Normal Random Variable*

Let

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{\frac{-x^2}{2}}$$

be the standard normal density, and define the cumulative normal distribution function

$$N(x) = \int_{-\infty}^{x} \varphi(\zeta) \, d\zeta$$

Note that this is strictly increasing and continuous, so it has a strictly increasing inverse $N^{-1}(y)$. We can construct an rv with a normal distribution on any probability space $(\Omega, \mathcal{F}, \mathbb{P})$ by starting with a uniformly distributed rv, $Y$ and setting $X = N^{-1}(Y)$. Then

$$\begin{aligned}\mu_X[a,b] &= \mathbb{P}\{\omega \in \Omega : a \leq X(\omega) \leq b\} \\ &= \mathbb{P}\{\omega \in \Omega : a \leq N^{-1}(Y(\omega)) \leq b\} \\ &= \mathbb{P}\{\omega \in \Omega : N(a) \leq Y(\omega) \leq N(b)\} \\ &= N(b) - N(a) \\ &= \int_a^b \varphi(x)\, dx\end{aligned}$$

This $\mu_X$ is called the **standard normal distribution**, and any rv with this distribution regardless of probability space is called a standard normal rv.

### 1.3. Expectations

We want to define a notion of the "average value" of an rv $X$ on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$. If $\Omega$ is finite, we can expect the "average" of the rv $X$ to be

$$\mathbb{E}X = \sum_{\omega \in \Omega} X(\omega)\mathbb{P}(\omega)$$

If $\Omega$ is countably infinite, we define $\mathbb{E}X$ as the same sum but infinite, but what about continous rvs? This requires the Lebesque integral, which I will not go into the construction of here (Analysis 3).

**Theorem 1** (Properties of the Lebesque Integral). Let $X$ be a random variable on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

(1) If $X$ takes on only finitely many values $y_1, y_2, \ldots, y_N$, then

$$\int_\Omega X(\omega)\; \mathbb{P}(\omega) = \sum_{k=1}^{N} y_k \mathbb{P}\{X = y_k\}$$

(2) (integrability) The r.v. $X$ is integrable iff $\int_\Omega |X(\omega)|\; d\mathbb{P}(\omega) < \infty$

(3) (comparison) If $X \leq Y$ almost surely,[1] and if $\int_\Omega X(\omega)\; d\mathbb{P}(\omega)$ and $\int_\Omega Y(\omega)\; d\mathbb{P}(\omega)$ are defined, then

$$\int_\Omega X(\omega)\; d\mathbb{P}(\omega) \leq \int_\Omega Y(\omega)\; d\mathbb{P}(\omega)$$

If $X = Y$ almost surely and these integrals are defined, then they are equal.

(4) (linearity) If $\alpha, \beta \in \mathbb{R}_{\geq 0}$, $X, Y$ nonnegative and integrable, then

$$\int_\Omega (\alpha X(\omega) + \beta Y(\omega))\; d\mathbb{P}(\omega) = \alpha \int_\Omega X(\omega)\; d\mathbb{P}(\omega) + \beta \int_\Omega Y(\omega)\; d\mathbb{P}(\omega)$$

*Remark.* Let $A \in \mathcal{F}$, then

$$\int_A X(\omega)\; d\mathbb{P}(\omega) = \int_\Omega \mathbb{I}_A(\omega) X(\omega)\; d\mathbb{P}(\omega)$$

**Definition 6** (expectation). Let $X$ be a random variable on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The **expectation** (or expected value) of $X$ is defined to be

$$\mathbb{E}X = \int_\Omega X(\omega)\; d\mathbb{P}(\omega)$$

This definition makes sense if $X$ is integrable, i.e. $\mathbb{E}|X| < \infty$ or $X \geq 0$ almost surely.

[1] i.e. $\mathbb{P}\{X \leq Y\} = 1$

Since the expectation is defined in terms of a Lebesque integral, all the previous properties mentioned for the Lebesque integral also hold for the expectation as well as **Jensen's inequality**, which states that if $\varphi$ is a convex, real-valued function defined on $\mathbb{R}$, and if $\mathbb{E}|X| < \infty$, then $\varphi(\mathbb{E}X) \leq \mathbb{E}\varphi(X)$.

**Example 6.** *Counting Heads*
Consider the infinite coin toss space defined previously, and let $\mathbb{P}$ be the probability measure that corresponds to probability $\frac{1}{2}$ for heads on each toss. Let

$$Y_n(\omega) = \begin{cases} 1 \text{ if } \omega_n = H \\ 0 \text{ if } \omega_n = T \end{cases}$$

Then

$$\mathbb{E}Y_n = 1 \cdot \mathbb{P}\{Y_n = 1\} + 0 \cdot \mathbb{P}\{Y_n = 0\} = \frac{1}{2}$$

**Example 7.** *Rationals and Irrationals*
Consider the probability space with $\Omega = [0, 1]$ and $\mathbb{P}$ the lebesque measure. Define

$$X(\omega) = \begin{cases} 1 \text{ if } \omega \text{ is irrational} \\ 0 \text{ if } \omega \text{ is rational} \end{cases}$$

Then

$$\mathbb{E}X = 1 \cdot \mathbb{P}\{\omega \in [0, 1] : \omega \text{ is irrational}\} + 0 \cdot \mathbb{P}\{\omega \in [0, 1] : \omega \text{ is rational}\}$$

Note there are only countably many rational numbers, and the probability of each individual one is 0, and thus by countable additivity $\mathbb{P}\{\omega \in [0, 1] : \omega \text{ is rational}\} = 0$. Therefore the probability of the irrationals must be 1, and subsequently $\mathbb{E}X = 1$.

It's worth noting that the lebesque integral is essential here, because the integral would be undefined with the Riemann definition.

**Definition 7** (Lebesque measure)**.** The **Lebesque measure** on $\mathbb{R}$, $\mathcal{L}$, is a measure that satisfies countable additivity and is defined as $\mathcal{L}[a, b] = b - a$ whenever $a \leq b$.

**Definition 8** (almost everywhere)**.** We say that a property holds **almost everywhere** (a.e.) if the set of numbers that fail to satisfy the property is of Lebesque measure 0.

**Theorem 2** (Riemann vs. Lebesque integrals)**.** Letting $f$ be a bounded function defined on $\mathbb{R}$, and $a < b$ in $\mathbb{R}$,

(1) The Riemann integrable $\int_a^b f(x)\,dx$ exists iff $f$ is continuous almost everywhere.

(2) If the Riemann integral is defined, then $f$ is Borel measurable[2] and the Riemann and Lebesque integrals are equal.

### 1.4. Convergence of Integrals

**Definition 9** (convergence of random variables)**.** Let $X_1, X_2, \ldots$ be a sequence of random variables, all defined on the same probability space. We say $\{X_k\}_{k\geq 0}$ **converges to** $X$ **almost surely** if

$$\mathbb{P}\left\{\omega \in \Omega : \lim_{k\to\infty} X_k(\omega) = X(\omega)\right\} = 1$$

[2] The preimage of every Borel set is a Borel set

**Example 8.** *Strong Law of Large Numbers*
Consider the infinite coin toss space with $\mathbb{P}$ chosen such that the probability of each head is $\frac{1}{2}$. Let

$$Y_k = \begin{cases} 1 \text{ if } \omega_k = H \\ 0 \text{ if } \omega_k = T \end{cases}$$

and

$$H_n = \sum_{k=1}^{n} Y_k$$

Then $H_n$ is the number of heads in the first $n$ tosses, and the strong law of large number asserts that

$$\lim_{n\to\infty} \frac{H_n}{n} = \frac{1}{2} \quad \text{almost surely}$$

Almost surely because there are some sequences where this isn't true, like all heads, but all those sequences have measure 0.

**Definition 10** (convergence of measurable functions)**.** If $\{f_k\}_{k\geq 1}$ is a sequence of real valued, Borel measurable functions on $\mathbb{R}$, then the sequence **converges to $f$ almost everywhere** if

$$\mathcal{L}\left\{x \in \mathbb{R} : \lim_{k\to\infty} f_k(x) \neq f(x)\right\} = 0$$

Almost everywhere and almost surely convergence is basically the same thing, just almost surely is used with r.v.s and probability spaces.

**Theorem 3** (monotone convergence)**.** If $\{X_n\}_{n\geq 0}$ is a sequence of almost surely montone increasing random variables that converge almost surely, then

$$\lim_{n\to\infty} \int_{-\infty}^{\infty} X_n(\omega)\, d\mathbb{P}(\omega) = \int_{-\infty}^{\infty} \lim_{n\to\infty} X_n(\omega)\, d\mathbb{P}(\omega)$$

Therefore $\lim_{n\to\infty} \mathbb{E}X_n = \mathbb{E}X$ where $\lim_{n\to\infty} X_n = X$.

Montone convergence theorem allows us to extend the expectation definition for the discrete case to countably infinite cases, getting that if $X$ takes on countably many values then $\mathbb{E}X = \sum_{k=0}^{\infty} x_k \mathbb{P}\{X = x_k\}$

**Example 9.** *St Petersburg paradox*
Same infinite coin toss space, define

$$X(\omega) = \begin{cases} 2 \text{ if } \omega_1 = H \\ 4 \text{ if } \omega_1 = T, \omega_2 = H \\ 8 \text{ if } \omega_1 = T, \omega_2 = T, \omega_3 = H \\ \vdots \\ 2^k \text{ if } \omega_1 = \omega_2 = \cdots = \omega_{k-1} = T, \omega_k = H \end{cases}$$

Then this is finite almost surely, since only non finite case is all tails. Then

$$\mathbb{E}X = \sum_{k\geq 1} 2^k \mathbb{P}\{X = 2^k\} = \sum_{k\geq 1} 2^k \frac{1}{2^k} = \infty$$

**Theorem 4** (dominated convergence)**.** If $\{X_n\}_{n\geq 0}$ is a sequence of random variables and there exists a rv $Y$ such that $\mathbb{E}Y < \infty$ and $|X_n| \leq Y$ almost surely for all $n$, then we can bring the limit inside the integral, similar to the MCT.

### 1.5. Computing Expectations

Computing expectations using the integral definition requires us to somehow turn it into an integral on $\mathbb{R}$. This can be done using a density function, but if we don't have one we can use the following. Note that the distribution measure is a probability measure on $\mathbb{R}$.

**Theorem 5.** $X$ a rv on probability space $(\Omega, \mathcal{F}, \mathbb{P})$, and $g$ a Borel measurable function on $\mathbb{R}$. Then

$$\mathbb{E}|g(X)| = \int_{\mathbb{R}} |g(x)| \, d\mu_X(x)$$

and if this quantity is finite then

$$\mathbb{E}g(X) = \int_{\mathbb{R}} g(x) \, d\mu_X(x)$$

*Proof.* The proof requires first showing that it is true for indicator functions, and then building up through simple functions then non-negative borel measurable functions and finally all Borel measurable functions. $\square$

We still can't compute this, but there are several cases we can. If $X$ takes only finitely many values then $\mu_X$ places a mass of size $p_k = \mathbb{P}\{X = x_k\}$ at each $x_k$ so we get the formula

$$\mathbb{E}g(X) = \int_{\mathbb{R}} g(x) \, d\mu_X(x) = \sum_{k=0}^{n} g(x_k) p_k$$

More commonly we have a nonnegative, Borel measurable density function $f$ on $\mathbb{R}$ such that

$$\mu_X(B) = \int_B f(x) \, dx \quad \text{for every Borel measurable set } B$$

and in the case when $f$ is bounded and a.e. continuous, we can compute the Riemann integral and compute the expectation.

**Theorem 6.** $X$ a random variable on $(\Omega, \mathcal{F}, \mathbb{P})$ and $g$ Borel measurable on $\mathbb{R}$. If $X$ has a density then

$$\mathbb{E}|g(X)| = \int_{-\infty}^{\infty} |g(x)| f(x) \, dx$$

and if this is finite then

$$\mathbb{E}g(X) = \int_{-\infty}^{\infty} g(x) f(x) \, dx$$

*Proof.* Once again, show true for indicator functions and build your way up to all Borel measurable functions. $\square$

### 1.6. Change of Measure

**Theorem 7.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a prob. space and $Z$ an almost surely nonnegative random variable with $\mathbb{E}Z = 1$. For $A \in \mathcal{F}$, define

$$\tilde{\mathbb{P}}(A) = \int_A Z(\omega) \, d\mathbb{P}(\omega)$$

Then $\tilde{\mathbb{P}}$ is a probability measure, and if $X$ is a nonnegative random variable then

$$\tilde{\mathbb{E}}X = \mathbb{E}[XZ]$$

If $Z$ is almost surely strictly positive then we also have

$$\mathbb{E}X = \tilde{\mathbb{E}}\left[\frac{X}{Z}\right]$$

*Remark.* For a rv with positive and negative values we can apply the theorem to its positive and negative parts provided the resulting subtraction does not result in $\infty - \infty$.

*Proof of theorem.* First

$$\tilde{\mathbb{P}}(\Omega) = \int_\Omega Z(\omega)\ d\mathbb{P}(\omega) = \mathbb{E}Z = 1$$

and for disjoint $A_1, A_2, \dots \in \mathcal{F}$, since $B_n = \cup_{k=1}^n A_k$ is increasing and $\lim_{n\to\infty} B_n = B_\infty = \cup_{k=1}^\infty A_k$ we get by monotone convergence theorem

$$\tilde{\mathbb{P}}\left(\bigcup_{k=1}^\infty A_k\right) = \int_\Omega \mathbb{I}_{B_\infty}(\omega)Z(\omega)\ d\mathbb{P}(\omega) = \lim_{n\to\infty}\int_\Omega \mathbb{I}_{B_n}(\omega)Z(\omega)\ d\mathbb{P}(\omega) = \sum_{k=1}^\infty \int_\Omega \mathbb{I}_{A_k}(\omega)Z(\omega)\ d\mathbb{P}(\omega) = \sum_{k=1}^\infty \tilde{\mathbb{P}}(A_k)$$

Thus it is a probability measure, also for the second part we prove for indicators and move our way up, here is the proof for indicators, assume $X = \mathbb{I}_A$

$$\tilde{\mathbb{E}}X = \tilde{\mathbb{P}}(A) = \int_\Omega \mathbb{I}_A(\omega)Z(\omega)\ d\mathbb{P}(\omega) = \mathbb{E}[\mathbb{I}_A Z] = \mathbb{E}[XZ]$$

□

**Definition 11.** Two probability measures on the same space $(\Omega, \mathcal{F})$ are said to be **equivalent** if they agree on which sets have probability zero.

The previously defined $\mathbb{P}$ and $\tilde{\mathbb{P}}$ are equivalent.

**Example 10.** Refer to example 4, our space is $[0,1]$, $\mathbb{P}$ is the Lebesque measure, and

$$\tilde{\mathbb{P}}[a,b] = \int_a^b 2\omega\ d\omega = b^2 - a^2, \quad 0 \le a \le b \le 1$$

Then since $\mathbb{P}$ is the lebesque measure, integrating with respect to $\omega$ is the same thing as integrating with respect to $\mathbb{P}(\omega)$, so we get

$$\tilde{\mathbb{P}}[a,b] = \int_a^b 2\omega\ d\mathbb{P}(\omega)$$

and since this holds for every closed interval it holds for every Borel set, and thus for $B \in \mathcal{B}$

$$\tilde{\mathbb{P}}(B) = \int_B 2\omega\ d\mathbb{P}(\omega)$$

Then with $Z(\omega) = 2\omega$,

$$\mathbb{E}Z = \int_0^1 2\omega\ d\omega = 1$$

So we can apply the previous theorem to any non-negative RV $X(\omega)$ and get

$$\int_0^1 X(\omega)\ d\tilde{\mathbb{P}}(\omega) = \tilde{\mathbb{E}}X = \mathbb{E}[XZ] = \int_0^1 X(\omega)\cdot 2\omega\ d\mathbb{P}(\omega)$$

and thus

$$d\tilde{\mathbb{P}}(\omega) = 2\omega\ d\mathbb{P}(\omega)$$

**Definition 12** (Radon-Nikodym derivative)**.** $(\Omega, \mathcal{F}, \mathbb{P})$ a prob. space and $\tilde{\mathbb{P}}$ another equivalent prob. measure on the same space, with an almost surely positive rv $Z$ such that $\tilde{\mathbb{P}}(A) = \int_A Z(\omega)\ d\mathbb{P}(\omega)$. Then $Z$ is called the **Radon-Nikodym derivative** of $\tilde{P}$ with respect to $\mathbb{P}$, and we write

$$Z = \frac{d\tilde{\mathbb{P}}}{d\mathbb{P}}$$

**Example 11.** *Change of Measure for a Normal RV*

Let $X$ be the standard normal rv defined on some prob. space $(\Omega, \mathcal{F}, \mathbb{P})$. $\Omega$ is uncountably infinite and thus $\mathbb{P}(\omega) = 0 \; \forall \omega \in \Omega$. Note $\mathbb{E}X = 0$ and $\text{Var}(X) = 1$. Define $Y = X + \theta$ for some constant $\theta$. Now $Y$ is a normal random variable with $\mathbb{E}Y = 1$. We want to perform a change of measure such that $\tilde{\mathbb{E}}Y = 0$, i.e. change the distribution without changing the actual random variable $Y$.

Define the random variable

$$Z(\omega) = e^{-\theta X(\omega) - \frac{1}{2}\theta^2} \quad \text{for all } \omega \in \Omega$$

Note $\mathbb{E}Z = 1$ and $Z(\omega) > 0 \; \forall \omega \in \Omega$ (I will not prove this is it easy to do). Now we can define

$$\tilde{\mathbb{P}}(A) = \int_A Z(\omega) \, d\mathbb{P}(\omega) \quad \forall A \in \mathcal{F}$$

We can then prove that $Y$ is the standard normal random variable with respect to $(\Omega, \mathcal{F}, \tilde{\mathbb{P}})$ by computing the cdf.

**Theorem 8** (Radon-Nikodym)**.** $\mathbb{P}$ and $\tilde{\mathbb{P}}$ equivalent probability measures on the same space $(\Omega, \mathcal{F})$. Then there exists an almost surely positive random variable $Z$ such that $\mathbb{E}Z = 1$ and for every $A \in \mathcal{F}$

$$\tilde{\mathbb{P}}(A) = \int_A Z(\omega) \, d\mathbb{P}(A)$$

### 1.7. Exercises

## 2. Information and Conditioning

### 2.1. Information and $\sigma$-algebras

**Definition 13** (filtration)**.** Let $\Omega$ be a nonempty set. Let $T$ be a fixed positive number and asume that for each $t \in [0, T]$ there is a $\sigma$-algebra $\mathcal{F}(t)$. Assume further that for every $s < t$, $\mathcal{F}(s) \subseteq \mathcal{F}(t)$. Then we call the collection of $\sigma$-algebras, $\mathcal{F}(t), \; t \in [0, T]$, a **filtration**.

**Example 12.**

Suppose $\Omega = C_0[0, T]$, the set of continuous functions on $[0, T]$ and that take the value zero at zero. Say we get to observe a specific $\tilde{\omega} \in \Omega$ up to time $0 \leq t \leq T$. Then certain subsets of $\Omega$ are resolved, like the subset $\{\omega \in \Omega : \max_{0 \leq s \leq t} \omega(s) \leq 1\}$ since we can exactly tell by time $t$ whether or not $\tilde{\omega}$ is in the set or not. This subset would thus go in $\mathcal{F}(t)$.

**Definition 14.** Let $X$ be a random variable defined on a nonempty space $\Omega$. Then the $\sigma$-algebra generated by $X$, denoted $\sigma(X)$, is the collection of all subsets of $\Omega$ of the form $\{X \in B\}$, where $B \in \mathcal{B}(\mathbb{R})$.

**Definition 15.** Let $X$ be a random variable defined on a non-empty sample space $\Omega$. Let $\mathcal{G}$ be a $\sigma$-algebra of subsets of $\Omega$. If every set in $\sigma(X)$ is also in $\mathcal{G}$, then we say that $X$ is $\mathcal{G}$-measurable.

A random variable is $\mathcal{G}$-measurable iff the information in $\mathcal{G}$ is sufficient to determine the value of $X$. Additionally, if $X$ is $\mathcal{G}$-measurable then so is $f(X)$ where $f$ is any Borel measurable function. If $X$ and $Y$ are $\mathcal{G}$-measurable then so it $f(X, Y)$ for any Borel measurable $f$, namely $X + Y$ and $XY$.

**Definition 16.** Let $\Omega$ be a nonempty sample space equipped with filtration $\mathcal{F}(t)$, $0 \leq t \leq T$. Let $X(t)$ be a collection of random variables indexed by $t \in [0, T]$. Then we say this collection of rvs is an **adapted stochastic process** if for each $t$ the random variable $X(t)$ is $\mathcal{F}(t)$ measurable.

### 2.2. Independence

Independence is when a random variable is independent of a $\sigma$-algebra, i.e. the $\sigma$-algebra gives no information about the value of the random variable. Independence is affected by the probability measure, whereas measurability is not affected. Independence of sets is something we have covered before, $A, B$ are independent if $\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$. We want to define independence of rvs similarly, in that if we know $X(\omega)$ then our estimation for the distribution of $Y$ is the same as it was before.

**Definition 17** (independence)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and $\mathcal{G}, \mathcal{H}$ two sub $\sigma$-algebras of $\mathcal{F}$. Then we say these two $\sigma$-algebras are independent if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B) \quad \forall A \in \mathcal{G}, B \in \mathcal{H}$$

Let $X$ and $Y$ be two random variables on this probability space. We say these random variables are independent if the two $\sigma$-algebras they generate, $\sigma(X)$ and $\sigma(Y)$, are independent. We say a random variable $X$ is independent of the $\sigma$-algebra $\mathcal{G}$ if $\sigma(X)$ and $\mathcal{G}$ are independent.

Note that the previous definition implies $X$ and $Y$ are independent iff for all Borel subsets $C, D \in \mathcal{B}(\mathbb{R})$

$$\mathbb{P}\{X \in C \text{ and } Y \in D\} = \mathbb{P}\{X \in C\} \cdot \mathbb{P}\{Y \in D\}$$

**Example 13.** Consider the stock price random variables based on the coin tosses defined before. The random variables $S_2$ and $S_3$ are intuitively not independent, but we can prove it by showing

$$p^3 = \mathbb{P}\{S_2 = 16 \text{ and } S_3 = 32\} = \mathbb{P}\{S_2 = 16\} \cdot \mathbb{P}\{S_3 = 32\} = p^5$$

Thus they are not independent, but $S_2$ and $\frac{S_3}{S_2}$ are independent because the latter only depends on the last coin toss.

Independence can be extended further than just two $\sigma$-algebras / rvs to any countable sequence, where the first $n$ $\sigma$-algebras / rvs are independent for all $n \geq 1$.

**Theorem 9.** Let $X$ and $Y$ be independent random variables, and $f$ and $g$ Borel-measurable functions on $\mathbb{R}$. Then $f(X)$ and $g(Y)$ are independent random variables.

*Proof.* You can prove this by demonstrating that $\sigma(f(X)) \subseteq \sigma(X)$ and $\sigma(g(Y)) \subseteq \sigma(Y)$ and then use their independence. □

**Definition 18** (joint distribution)**.** Let $X$ and $Y$ be two random variables. The pair of random variables $(X, Y)$ takes values in the plane $\mathbb{R}^2$ and the **joint distribution measure** of $(X < Y)$ is given by

$$\mu_{X,Y}(C) = \mathbb{P}\{(X, Y) \in C\} \quad \forall C \in \mathcal{B}(\mathbb{R}^2)$$

Similarly we can define a joint cdf

$$F_{X,Y}(a, b) = \mu_{X,Y}((-\infty, a] \times (-\infty, b]) = \mathbb{P}\{X \leq a, Y \leq b\}$$

and joint density $f_{X,Y}(x, y)$ such that

$$\mu_{X,Y}(C) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \mathbb{I}_C(x, y) f_{X,Y}(x, y)\, dy\, dx$$

for all Borel sets $C \subset \mathbb{R}^2$. This holds iff

$$F_{X,Y}(a, b) = \int_{-\infty}^{a} \int_{-\infty}^{b} f_{X,Y}(x, y)\, dy\, dx$$

for all $a, b \in \mathbb{R}$.

The marginal distribution measures of $X$ and $Y$ are

$$\mu_X(A) = \mathbb{P}\{X \in A\} = \mu_{X,Y}(A \times \mathbb{R}), \ \mu_Y(B) = \mathbb{P}\{Y \in B\} = \mu_{X,Y}(\mathbb{R} \times B)$$

You can also get the marginal densities by integrating out the other variable.

**Theorem 10.** Let $X$, $Y$ be random variables. The following are equivalent:

(1) $X$ and $Y$ are independent.

(2) $\mu_{X,Y}(A \times B) = \mu_X(A) \cdot \mu_Y(B)$ for all $A, B \in \mathcal{B}(\mathbb{R})$.

(3) $F_{X,Y}(a,b) = F_X(a) \cdot F_Y(b)$ for all $a, b, \in \mathbb{R}$.

(4) $\mathbb{E}e^{uX+vY} = \mathbb{E}e^{uX} \cdot \mathbb{E}e^{vY}$ for $u, v \in \mathbb{R}$ s.t. the expectations are finite.

(5) $f_{X,Y}(x,y) = f_X(x) \cdot f_Y(y)$ for a.e. $x, y \in \mathbb{R}$ if a density exists.

The above also imply that $\mathbb{E}[XY] = \mathbb{E}X \cdot \mathbb{E}Y$ provided $\mathbb{E}|XY| < \infty$.

*Proof.* Not too difficult just rearranging defns, exercise. ☐

**Example 14.** *Independent Normal RVs*
$X$ and $Y$ are independent and standard normal if they have joint density

$$f_{X,Y}(x,y) = \frac{1}{2\pi} e^{-\frac{1}{2}(x^2+y^2)}$$

This is imply the product of the standard normal density for $X$ and $Y$.

**Definition 19** (variance, covariance)**.** Let $X$ be a rv with $\mathbb{E}|X| < \infty$. The **variance** of $X$, denoted $\operatorname{Var}(X)$ is defined

$$\operatorname{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2]$$

The standard deviation of $X$ is $\sqrt{\operatorname{Var}(X)}$. Additionally by linearity of expectation we have

$$\operatorname{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}X)^2.$$

Letting $Y$ be another rv and assuming $\mathbb{E}X, \mathbb{E}Y, \operatorname{Var}(X), \operatorname{Var}(Y)$ are all finite we define the **covariance** of $X$ and $Y$ to be

$$\operatorname{Cov}(X,Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)]$$

Again by linearity we get

$$\operatorname{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}X \cdot \mathbb{E}Y$$

Assume additionally that $\operatorname{Var}(X), \operatorname{Var}(Y) > 0$, then the **correlation coefficient** of $X$ and $Y$ is defined

$$\rho(X,Y) = \frac{\operatorname{Cov}(X,Y)}{\sqrt{\operatorname{Var}(X)\operatorname{Var}(Y)}}$$

If $\rho(X,Y) = 0$ we say that $X, Y$ are uncorrelated.

### 2.3. Conditional Expectations

**Definition 20.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a prob space, $\mathcal{G}$ a sub $\sigma$-algebra of $\mathcal{F}$ and $X$ a rv that is either non-negative or integrable. The conditional expectation of $X$ given $\mathcal{G}$, denoted $\mathbb{E}[X|\mathcal{G}]$, is any random variable that satisfies

(1) (measurability) $\mathbb{E}[X|\mathcal{G}]$ is $\mathcal{G}$ measurable. [3]

(2) (partial averaging)

$$\int_A \mathbb{E}[X|\mathcal{G}](\omega)\, d\mathbb{P}(\omega) = \int_A X(\omega)\, d\mathbb{P}(\omega) \quad \text{for all } A \in \mathcal{G}$$

[3] If $\mathcal{G}$ is the $\sigma$-algebra generated by some random variable $W$, then we write $\mathbb{E}[X|W]$ instead of $\mathbb{E}[X|\sigma(W)]$ .

*Proof of Uniqueness.* Assume for contradiction it is not unique, and there exists two rvs $Z$ and $Y$ that satisfy the previous definition. They are both then $\mathcal{G}$ measurable, and thus their difference also is and thus the set $A = \{Y - Z > 0\}$ is in $\mathcal{G}$. By property (2) of the definition, we also have

$$\int_A (Y(\omega) - Z(\omega))\, d\mathbb{P}(\omega) = 0$$

The only way this integral could be zero is if $A$ has probability zero (since the integrand is strictly positive) and thus $Y \leq Z$ almost surely. Reversing $Y$ and $Z$ in this argument we also get $Z \leq Y$ almost surely, and thus $Y = Z$ almost surely. □

**Theorem 11.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\mathcal{G}$ a sub $\sigma$-algebra of $\mathcal{F}$. The following are properties of conditional expectations.

(1) (Linearity) If $X$ and $Y$ are integrable rvs and $c_1, c_2$ are constants then

$$\mathbb{E}[c_1 X + c_2 Y | \mathcal{G}] = c_1 \mathbb{E}[X|\mathcal{G}] + c_2 \mathbb{E}[Y|\mathcal{G}]$$

(2) If $X$ and $Y$ are integrable rvs, $Y$ and $XY$ are integrable, and $X$ is $\mathcal{G}$ measurable, then

$$\mathbb{E}[XY|\mathcal{G}] = X\mathbb{E}[Y|\mathcal{G}]$$

This also holds if $X$ is positive and $Y$ is nonnegative, but might be $\pm\infty$.

(3) (Iterated Conditioning) If $\mathcal{H}$ is a sub $\sigma$-algebra of $\mathcal{G}$ (i.e. contains less information) and $X$ is an integrable rv, then

$$\mathbb{E}[\mathbb{E}[X|\mathcal{G}]|\mathcal{H}] = \mathbb{E}[X|\mathcal{H}]$$

(4) (Independence) If $X$ is an integrable rv and independent of $\mathcal{G}$, then

$$\mathbb{E}[X|\mathcal{G}] = \mathbb{E}X$$

This also holds if $X$ is nonnegative, but $\mathbb{E}X$ might be $\pm\infty$.

(5) (Conditional Jensen's Inequality) If $\varphi(x)$ is a convex function of a dummy variable $x$, and $X$ is an integrable rv, then

$$\mathbb{E}[\varphi(X)|\mathcal{G}] \geq \varphi(\mathbb{E}[X|\mathcal{G}])$$

*Proof.*

(1) The right and side is obviously $\mathcal{G}$ measurable since the components are $\mathcal{G}$ measurable, and thus we can just prove that the right side satisfies the partial averaging, i.e.

$$\int_A (c_1 \mathbb{E}[X|\mathcal{G}](\omega) + c_2 \mathbb{E}[Y|\mathcal{G}](\omega))\, d\mathbb{P}(\omega) = \int_A (c_1 X(\omega) + c_2 Y(\omega))\, d\mathbb{P}(\omega)$$

(2) Again, similar to (1) the right side is $\mathcal{G}$ measurable and we just have to prove the partial averaging condition. This requires building up from $X$ being an indicator to simple function, etc...

(3) The right side is obviously $\mathcal{H}$ measurable, so we just have to show it satisfies the partial averaging. $\forall A \in \mathcal{H}$,

$$\int_A \mathbb{E}[\mathbb{E}[X|\mathcal{G}]|\mathcal{H}](\omega)\, d\mathbb{P}(\omega) = \int_A \mathbb{E}[X|\mathcal{G}](\omega)\, d\mathbb{P}(\omega)$$

Additionally $\forall A \in \mathcal{H}$,

$$\int_A \mathbb{E}[X|\mathcal{H}](\omega)\, d\mathbb{P}(\omega) = \int_A X(\omega)\, d\mathbb{P}(\omega) = \int_A \mathbb{E}[X|\mathcal{G}](\omega)\, d\mathbb{P}(\omega)$$

(4) $\mathbb{E}X$ is not random so it is measurable w.r.t. any $\sigma$-algebra, including $\mathcal{G}$. Thus we just have to show the partial averaging property, which requires building $X$ up from indicator to simple function etc...

□

**Lemma 12** (Independence)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a prob space and $\mathcal{G}$ a sub $\sigma$-algebra of $\mathcal{F}$. Then suppose the rvs $X_1, \dots, X_K$ are $\mathcal{G}$ measurable and the rvs $Y_1, \dots Y_L$ are independent of $\mathcal{G}$. Let $f(x_1, \dots, x_K, y_1, \dots, y_L)$ be a function and define

$$g(x_1, \dots, x_K) = \mathbb{E}f(x_1, \dots, x_K, Y_1, \dots, Y_L)$$

Then

$$\mathbb{E}[f(X_1, \dots, X_K, Y_1, \dots, Y_L)|\mathcal{G}] = g(X_1, \dots, X_K)$$

**Definition 21** (martingales)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a prob space, $T$ a fixed positive number, and let $\mathcal{F}(t)$, $0 \le t \le T$, be a filtration of sub $\sigma$-algebras of $\mathcal{F}$. Consider an adapted stochastic process $M(t), 0 \le t \le T$.

(1) If

$$\mathbb{E}[M(t)|\mathcal{F}(s)] = M(s) \quad \forall\, 0 \le s \le t \le T$$

then we say this process is a **martingale**. It has no tendency to rise or fall.

(2) If

$$\mathbb{E}[M(t)|\mathcal{F}(s)] \ge M(s) \quad \forall\, 0 \le s \le t \le T$$

then we say this process is a **submartingale**. It has no tendency to fall, and may have a tendency to rise.

(3) If

$$\mathbb{E}[M(t)|\mathcal{F}(s)] \le M(s) \quad \forall\, 0 \le s \le t \le T$$

then we say this process is a **supermartingale**. It has no tendency to rise, and may have a tendency to fall.

**Definition 22** (Markov process)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a prob space, $T$ a fixed positive number, and let $\mathcal{F}(t)$, $0 \le t \le T$, be a filtration of sub $\sigma$-algebras of $\mathcal{F}$. Consider an adapted stochastic process $X(t), 0 \le t \le T$. Assume that for all $0 \le s \le t \le T$ and for every non negative Borel-measurable function $f$, there is another Borel-measurable function $g$ such that

$$\mathbb{E}[f(X(t))|\mathcal{F}(t)] = g(X(s))$$

Then we say that $X$ is a **Markov process**.

### 2.4. Exercises

**Example 15.** *Problem 1*

$(\Omega, \mathcal{F}, \mathbb{P})$ a prob space, and rv $X$ is measurable with respect to the trivial $\sigma$-algebra, $\mathcal{F}_0 = \{\emptyset, \Omega\}$. This implies that $\sigma(X) \subseteq \mathcal{F}_0$. Thus $\{\omega \in \Omega : X(\omega) \in B\} \in \{\emptyset, \Omega\}, \forall B \in \mathcal{B}_{\mathbb{R}}$.

Suppose for contradiction $X$ is not a constant. Then there exists a $\omega_1, \omega_2 \in \Omega$ such that $X(\omega_1) \neq X(\omega_2)$. Let $a = X(\omega_1)$. Then $\{a\}$ is a Borel set and $\omega_1 \in X^{-1}(\{a\}) \in \{\emptyset, \Omega\}$ thus it is non empty and must be equal to $\Omega$. Then, $\forall \omega \in \Omega, X(\omega) = a$ which is a contradiction to $X(\omega_2) \neq X(\omega_1) = a$. Therefore $X$ must be constant.

**Example 16.** *Problem 2*

(1) $\sigma(X) = \{\{X \in B\}\ \forall B \in \mathcal{B}_{\mathbb{R}}\} = \{\emptyset, \{S_2 = 4\}, \{S_2 \neq 4\}, \Omega_2\} = \{\emptyset, \{HT, TH\}, \{TT, HH\}, \Omega_2\}$

(2) $\sigma(S_1) = \{\{S_1 \in B\}\ \forall B \in \mathcal{B}_{\mathbb{R}}\} = \{\emptyset, \{HH, HT\}, \{TH, TT\}, \Omega_2\}$

(3) These $\sigma$-algebras are independent if $\tilde{\mathbb{P}}(A \cap B) = \tilde{\mathbb{P}}(A) \cdot \tilde{\mathbb{P}}(B) \quad \forall A \in \sigma(X), \forall B \in \sigma(S_1)$. This is obviously true if $A = \emptyset, \Omega_2$ or $B = \emptyset, \Omega_2$ so we only check intermediate cases. This can be proved with some brute force which I will not do here.

(4) Consider $A = \{HH, HT\}, B = \{HT, TH\}$. $\mathbb{P}(A) = \frac{2}{3}, \mathbb{P}(B) = \frac{4}{9}$. Thus $\mathbb{P}(A) \cdot \mathbb{P}(B) = \frac{8}{27}$. But $\mathbb{P}(A \cap B) = \mathbb{P}(\{HT\}) = \frac{2}{9} = \frac{6}{27}$, thus they are not equal.

## 3. Brownian Motion

### 3.1. Scaled Random Walks

This section is more about building intuition and less about specifics.

We begin with a symmetric random walk. To construct it, we repeatedly toss a fair coin and denote the successive outcomes as $\omega = \omega_1\omega_2\omega_3 \dots$. Let

$$X_j = \begin{cases} 1 & \text{if } \omega_j = H \\ 0 & \text{if } \omega_j = T \end{cases}$$

and define $M_0 = 0$,

$$M_k = \sum_{j=1}^{k} X_j, \quad k = 1, 2, \dots$$

The process $M_k,\ k = 1, 2, \dots$ is a **symmetric random walk**.

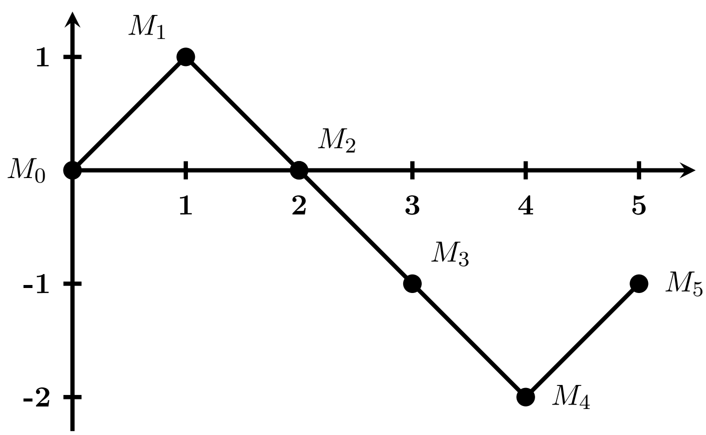


Figure 1: **Five steps of a random walk.**

For any nonnegative integers $0 = k_0 < k_1 < k_2 < \cdots < k_m$ the random variables

$$M_{k_1} = (M_{k_1} - M_{k_0}), (M_{k_2} - M_{k_1}) \dots, (M_{k_m} - M_{k_{m-1}})$$

are independent and

$$M_{k_{i+1}} - M_{k_i} = \sum_{j=k_{i+1}}^{k_{i+1}} X_j$$

is called the **increment** of the random walk. Increments have expectation 0 and are independent if they do not overlap. Additionally, the variance of $M_{k_{i+1}} - M_{k_i}$ is $k_{i+1} - k_i$.

The symmetric walk is a martingale.

*Proof.*

$$\begin{aligned}\mathbb{E}[M_\ell|\mathcal{F}_k] &= \mathbb{E}[(M_\ell - M_k) + M_k|\mathcal{F}_k]\\ &= \mathbb{E}[M_\ell - M_k|\mathcal{F}_k] + \mathbb{E}[M_k|\mathcal{F}_k]\\ &= \mathbb{E}[M_\ell - M_k|\mathcal{F}_k] + M_k\\ &= \mathbb{E}[M_\ell - M_k] + M_k = M_k\end{aligned}$$

□

The **quadratic variation** of the symmetric random walk up to time $k$ is defined to be

$$[M, M]_k = \sum_{j=1}^{k}(M_j - M_{j-1})^2 = k$$

Note this is the same as the variance, but the variance is an average over all paths so it takes the probability of heads vs tails into account, whereas the quadratic variation is computed along a single path. Since it only requires one path, it can be calculated explicitly in practice whereas variance cannot. For a random walk the path chosen doesn't affect the quadratic variation, but for other processes it might.

We define the **scaled symmetric walk** by fixing a positive integer $n$ and defining

$$W^{(n)}(t) = \frac{1}{\sqrt{n}}M_{nt}$$

provided that nt is an integer. Otherwise $W^{(n)}(t)$ is the linear interpolation between the nearest values where $nt$ is an integer.

The scaled random walk acts just like the regular random walk but with much finer time increments (n times to be exact). So, it will have the same expected value and variance at the same times as the regular random walk.

The quadratic variation of the scaled random walk can be computed just like the standard random walk

$$[W^{(n)}, W^{(n)}]_t = \sum_{j=1}^{nt}\left(W^{(n)}\left(\frac{j}{n}\right) - W^{(n)}\left(\frac{j-1}{n}\right)\right)^2 = \sum_{j=1}^{nt}\left[\frac{1}{\sqrt{n}}X_j\right]^2 = t$$

**Theorem 13** (Central Limit)**.** As $n$ goes to $\infty$, the distribution of the scaled random walk $W^{(n)}(t)$ converges to the normal distribution with mean 0 and variance $t$.

*Proof.* This can be proved using the moment generating function of the normal distribution and for the scaled random walk and showing that they are equal. Refer to the book for the proof, but it's just a lot of rearranging etc... □

Now we can use this theorem to prove the following. Assume we had a binomial asset pricing model on time interval $[0, t]$ such that we have chosen an integer $n$ and $nt$ is an integer. The model takes $n$ steps per unit of time. Assume the up factor and down factor are $u_n = 1 + \frac{\sigma}{\sqrt{n}}$ and $d_n = 1 - \frac{\sigma}{\sqrt{n}}$ respectively. Then the risk neutral probabilities are [4]

$$\tilde{p} = \frac{1 + r - d_n}{u_n - d_n} = \frac{\frac{\sigma}{\sqrt{n}}}{2\frac{\sigma}{\sqrt{n}}} = \frac{1}{2} \quad \text{and} \quad \tilde{q} = \frac{u_n - 1 - r}{u_n - d_n} = \frac{\frac{\sigma}{\sqrt{n}}}{2\frac{\sigma}{\sqrt{n}}}$$

The stock price at time $t$ is determined by the initial stock price and result of the first $nt$ coin tosses, and using the fact that $M_{nt} = H_{nt} - T_{nt}$ and $nt = H_{nt} + T_{nt}$ we can then get

$$S_n(t) = S(0)u_n^{H_{nt}}d_n^{T_{nt}} = S(0)\left(1 + \frac{\sigma}{\sqrt{n}}\right)^{\frac{1}{2}(nt+M_{nt})}\left(1 - \frac{\sigma}{\sqrt{n}}\right)^{\frac{1}{2}(nt-M_{nt})}$$

[4] Assuming an interest rate $r = 0$

**Theorem 14.** As $n \to \infty$, the distribution of $S_n(t)$ converges to the distribution[5] of

$$S(t) = S(0)e^{\sigma W(t) - \frac{1}{2}\sigma^2 t}$$

*Proof.* Take the log of the $S_n(t)$ equation and use the taylor expansion of $\log(1+x) = x - \frac{1}{2}x^2 + \mathcal{O}(x^3)$. Then

$$\log S(t) = \log S(0) + \frac{1}{2}(nt + M_{nt})\left(\frac{\sigma}{\sqrt{n}} - \frac{\sigma^2}{2n} + \mathcal{O}(n^{-3/2})\right) + \frac{1}{2}(nt - M_{nt})\left(-\frac{\sigma}{\sqrt{n}} - \frac{\sigma^2}{2n} + \mathcal{O}(n^{-3/2})\right)$$

With some simplification this gets us what we need. □

## 3.2. Brownian Motion

[5] This distribution is called log-normal, in this case with mean $-\frac{1}{2}\sigma^2 t$ and variance $\sigma^2 t$